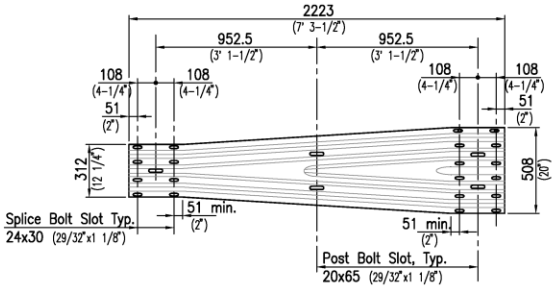
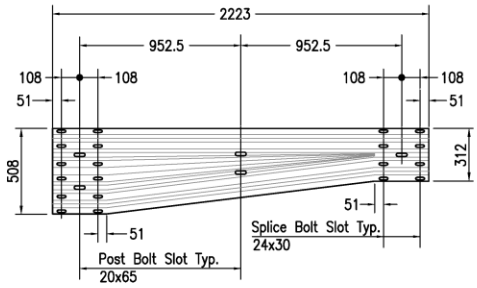
Steel Guardrail Beam Parts

SP312–7

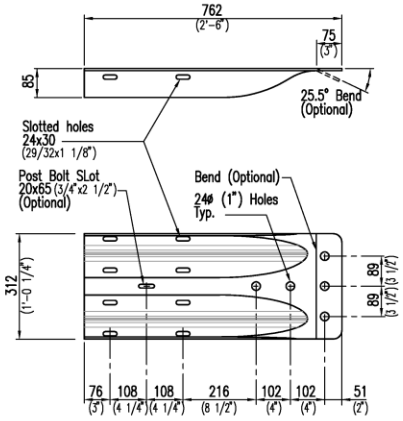
W – Thrie Beam Transition Section RWT01a–b



W – Thrie Beam Transition Section RWT02a–b



W Beam Terminal Connector RWE02b



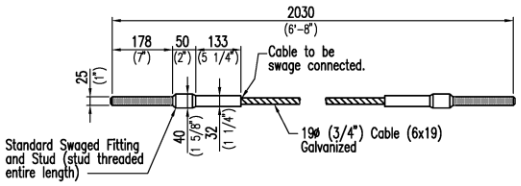
NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

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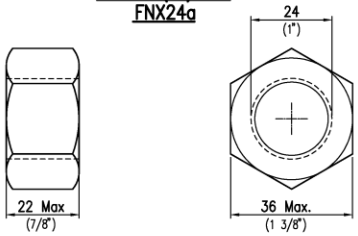
Steel Guardrail – B.C.T. End Assembly Parts

SP312–8

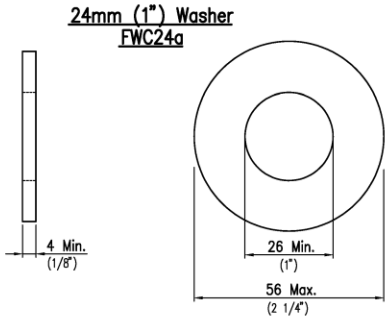
BCT Cable Anchor Assembly
FCA01



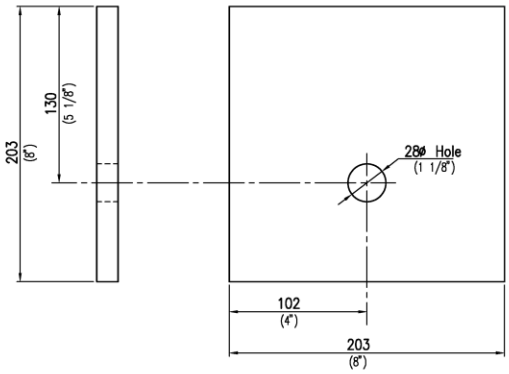
24mm (1") Nut
FNX24a



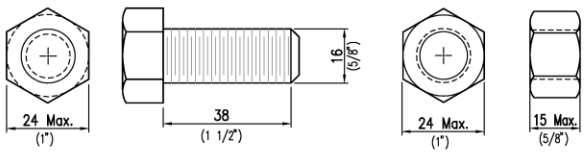
24mm (1") Washer
FWC24a



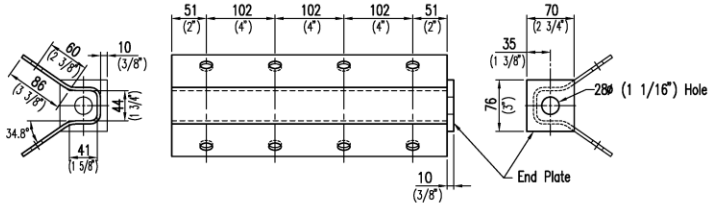
BCT (Timber) Bearing Plate
FPB01



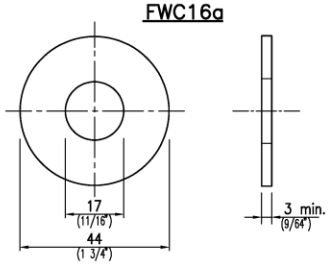
16mm (5/8") Hex Bolt and Nut
FBX16a



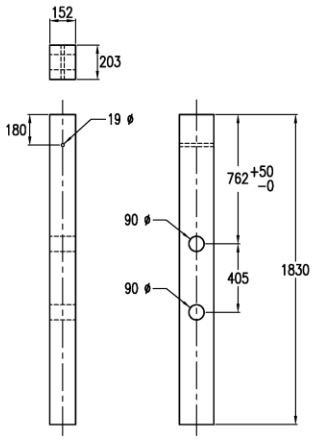
BCT Anchor Plate
FPA01



Washer for 16mm (5/8") Bolt
FWC16a



CRT Post

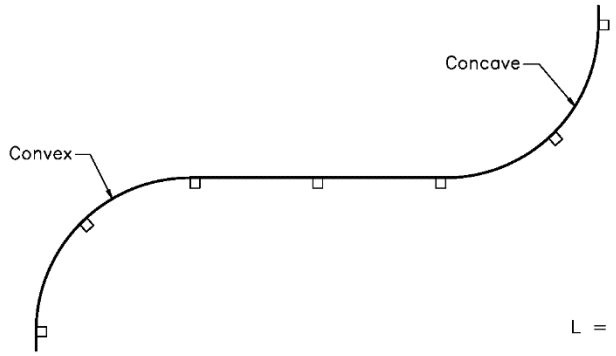


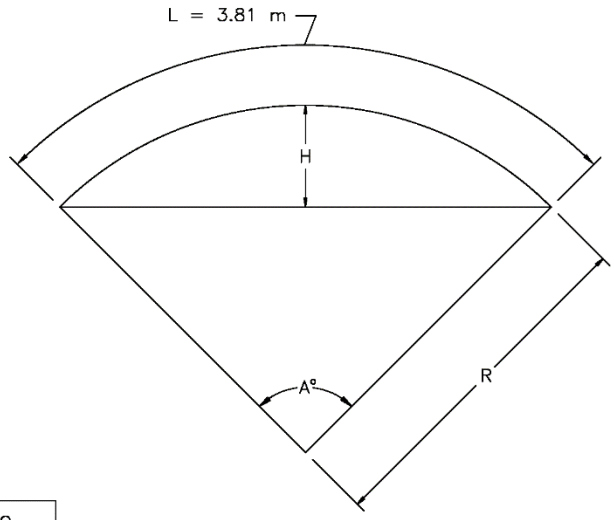
NOT TO SCALE
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Curved Rails

SP312-11





Typical Radii R (m)	Approximate A (deg)	Approximate H (mm)
1.5 (min. convex)	145	1055
3.0 (min. concave)	72	585
5.0	43	360
7.5	29	240
10.0	21	180
12.5	17	145
15.0	14	120
18.0	12	100
21.0	10	85
25.0	8	70
30.0	7	60
35.0	6	50
40.0	5	45
45.0	4.5	40

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SECTION 316

STEEL AND WIRE FENCE MATERIALS

316.01 Scope – This Section covers the quality and manufacture of wire fencing material, steel posts, gates and hardware, and various accessory materials and the fabrication of steel tubular fencing for pedestrian and bicyclist applications.

Treated wood fence posts are specified by SS 909.

For convenience, chain link fabric necessary for protection on rock slopes is included in this Section.

The uses of materials specified in this Section are generally indicated on Drawings of the SP741 series for construction of fences and SS 207 for rock protection.

The types of Standard Wire fencing and Chain Link fabric for fencing and protection on rock slopes covered by this Section are designated as follows:

- **Type A** Special Wire Fabric Fence for use only on railway rights-of-way
- **Type B** Standard Wire Fabric Fence
- **Type C** Standard Barbed Wire Fence
- **Type D** Chain Link Fence.

Chain link fence protection on rock slopes and tentative requirements for High-tensile Smooth-wire Fences are also included.

316.02 Applicable Documents – See Table 316-A.

Note: Standards referred to in this Section are obtainable from:

CGSB: Publishing and Depository Services
Public Works and Government Services Canada
Ottawa ON K1A 0S5
https://www.tpsgc-pwgsc.gc.ca/ongc-cgsb/boutique-store-eng.html

Table 316-A: Documents Applicable for Reference

ASTM STANDARD (unless otherwise noted)	Specification title
CAN/CGSB138.1-M	Fabric for Chain Link Fence
CSA C83	Communication and Power Line Hardware [for Strain Age Embrittlement test]
A53	Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated Welded and Seamless
A90	Standard Test Method for Weight [Mass] of Coating on Iron or Steel Articles with Zinc or Zinc-Alloy Coatings
A116	Standard Specification for Metallic-Coated Carbon Steel Woven Wire Fence Fabric
A121	Standard Specification for Metallic-Coated Carbon Steel Barbed Wire
A123	Standard Specification for Zinc (Hot-Dipped Galvanized) Coatings on Iron and Steel Products
A153	Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
A370	Standard Test Methods and Definitions for Mechanical Testing of Steel Products
A641	Standard Specification for Zinc-Coated (Galvanized) Carbon Steel Wire
A1011	Standard Specification for Steel, Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength
B6	Standard Specification for Zinc (Slab Zinc)
B117	Standard Practice for Operating Salt Spray (Fog) Apparatus
B211	Specification for Aluminum-Alloy Bar, Rod, and Wire
D2247	Standard Practice for Testing Water Resistance of Coatings in 100 % Relative Humidity
E4	Standard Practices for Force Verification of Testing Machines
F1043	Standard Specification for Strength and Protective Coatings on Steel Industrial Fence Framework
G152	Standard Practice for Operating Open Flame Carbon Arc Light Apparatus for Exposure of Nonmetallic Materials
G155	Standard Practice for Operating Xenon Arc Light Apparatus for Exposure of Non-Metallic Materials

ASTM: American Society for Testing and Materials
100 Barr Harbor Drive
West Conshohocken, PA
19428-2959
USA
http://www.astm.org/

316.03 Type A & B Fence Fabrics – Fence fabric for railway and highway rights-of-way shall be manufactured from good commercial quality galvanized steel wire in general accordance with ASTM A116, but shall conform to the requirements following and those listed in Table 316-B.

316.03.01 Styles – See Table 316-B. Length of fabric rolls shall be 101 m unless otherwise ordered.

Note: Other standard and specialty fabrics when specified shall meet the following general specifications.

316.03.02 Wire Size and Strength – All wire used in the fabric shall have a minimum galvanized coated nominal diameter of 2.50 mm (12 1/2 ga.). The variation permitted from the measured nominal diameter shall be $\pm$ 0.1 mm. The minimum breaking force shall be 4250 N.

316.03.03 Zinc Coating – The zinc used for coating shall be any grade of zinc conforming to ASTM B6. All wire shall have a uniform coating of zinc which shall adhere firmly to the base metal. The average mass of zinc coating shall be 90 g/m^2 of uncoated wire surface for three specimens but minimum 80 g/m^2 for any individual specimen.

316.03.04 Quality of Work – Splicing of individual line wires by means of a wrap joint, electric butt weld or galvanized sleeve is permitted.

The maximum number of line wire joints shall not exceed half of the number of line wires in any roll of fencing.

All joints shall be made in accordance with good work practice.

All stay wires shall be properly spaced and substantially perpendicular to the line wires.

The locks or fastenings at the intersections of the longitudinal and vertical wires of the woven wire fencing shall be of such construction as will prevent them from slipping, either longitudinally or vertically. All locks and fastenings shall be effectively galvanized.

Horizontal wires of all woven wire fencing shall be provided with tension curves to accommodate temperature changes after erection.

316.03.05 Testing

(a) Samples – Sufficient length of fabric to include at least three vertical stay wires shall be cut from each roll to be tested as required by SS 316.15from which line wire and stay wire specimens shall be taken for testing.

(b) Diameter of Coated Wire – The average of two readings taken at right angles to each other at random on uniformly coated areas shall all be within the tolerance measurements specified in SS 316.03.02.

(c) Strength – The tensile strength of three line wire specimens shall be tested in accordance with ASTM A370 on a machine calibrated to ASTM E4 standards. The tensile strengths shall meet the requirements of SS 316.03.02.

(d) Coating Adhesion – Three line wire specimens shall be wrapped in a close helix at least twice around a cylindrical mandrel having a diameter four times the nominal diameter of the wire: wrapping speed shall not exceed four seconds per turn. The coating shall adhere to the wire without flaking or being removed by rubbing with bare fingers.

(e) Coating Mass – Three line wire specimens (from each roll), each more than 400 mm in length, and three stay wire specimens of short lengths between knots, etc., but each specimen totalling more than 400 mm shall be tested in accordance with ASTM A90 for conformity with the requirements of SS 316.03.03.

(f) Rejection and Retest: Refer to SS 316.16.

316.04 Type D Fence Chain Link Fabric – Chain link fence fabric shall be manufactured from good commercial quality galvanized steel wire (or other specified finish) in general accordance with CAN/CGSB-138.1M and shall conform to the requirements listed in SS 316.04.01 through SS 316.04.04.

316.04.01 Fabric Classification (according to CGSB Standard) – The type, class, style and grade of chain link fence fabric shall be as specified on the Purchase Order, Work Order or Special Provisions in accordance with the Drawings and/or SS Drawings SP741-05.01 and SP741-05.02, selected from the following or as may be otherwise specified or approved.

(a) Type/Class 1A – Zinc coated (galvanized) steel wire fabric normally will be specified for all heights of chain link fences as indicated on Drawing

Table 316-B: Styles of Fence

Fence Types	Fabric Style No.	Number of Horizontal Wires	Fabric Height (mm)	Stay Wire Spacing (mm)	Wire Diameter (mm)
A	1047-6	10	1195	150	2.50 min.
B	939-6	9	990	150	2.50 min.

SP741-05.01, unless Type/Class 1B Vinyl coated steel is required for authorized special installations.

(b) Style – nominal diameter of zinc coated wire:

(i) Heavy, for highway protection fence from rolling rock

(ii) Medium, for Standard and Security height fences

(iii) Light, for Residential height fences (see SS Drawing SP741-05.01)

(iv) See Table 316-C and Table 316-D.

(c) Grade – mass of zinc coating per unit area of uncoated wire: Grade 1, average 490 g/m^2 for three specimens, minimum 440 g/m^2 for any individual specimen. The zinc used for coating shall be any grade of zinc conforming to ASTM B6. All wire shall have a uniform coating of zinc which shall adhere firmly to the base material.

Note: When specified, Grade 2 heavy coating (average 610 g/m^2, individual 550 g/m^2 minimums) for corrosive environments and the zinc coating (average 90 g/m^2, individual 80 g/m^2 minimums) for vinyl coated wire and the vinyl coating requirements shall be in accordance with CAN/CGSB-138.1M.

316.04.02 Fabric Height, Length and Mesh – Wire shall be woven throughout to a uniform diamond mesh 50 mm ± 3.0 mm clear between parallel mesh wires: fabric height as specified ± 25 mm with both selvage edges twisted and/or knuckled:

(a) fabric 1.5 m high and under: knuckled both edges

(b) fabric 1.8 m high and over: knuckled one edge, and twisted the other

(c) Residential height fence light style fabric 1.0 m and 1.2 m high: knuckled both edges.

Length of rolls with slack removed by nominal tensioning shall be 100 m unless otherwise ordered.

Note: Typical (normal) ordering designation from above: Type 1, Class A, Grade 1, Medium Style with roll width (height), length, mesh size and quantity.

316.04.03 Quality of Work – Fabric shall be of uniform quality and condition, free from all defects and imperfections affecting serviceability and appearance.

316.04.04 Testing

(a) Samples – Three full height pickets shall be provided from each roll as required by SS 316.15.

(b) Diameter of Coated Wire – The average of two readings taken at right angles to each other at random on uniformly coated portions of each pair of pickets between bends shall be within the tolerance measurement specified in SS 316.04.01(b).

(c) Strength – The tensile strength of three line wire specimens shall be tested in accordance to ASTM A370 on a machine calibrated to ASTM E4 standards. The tension applied to the undeformed portion between two adjacent bends shall meet the requirements of SS 316.04.01(b).

(d) Coating Adhesion – Three specimens of undeformed portions between two adjacent bends shall be wrapped in a close helix once around a cylindrical mandrel having a diameter four times the nominal diameter of the wire: wrapping speed shall not exceed four seconds per turn The coating shall adhere to the wire without flaking or being removed by rubbing with bare fingers.

(e) Coating mass – Three specimens (from each roll) each more than 400 mm long excluding knuckled or twisted ends shall be tested in accordance with ASTM A90 for conformity with the requirements of SS 316.04.01(c).

Table 316-C: Zinc Coated Wire

Type 1A	Style	Zinc Coated Wire Nominal Diameter	Breaking Force Minimum (N)
	Heavy	5.0 mm ± 0.1 mm	10 000
	Medium	3.5 mm ± 0.1 mm	5 000
	Light	3.0 mm ± 0.1 mm	3 600

Table 316-D: Vinyl Coated Zinc-Coated Wire (When Permitted)

Type 1B	Style	Vinyl Coated Zinc-Coated Wire Nominal Diameter	Breaking Force Minimum (N)
	Heavy	5.0 mm ± 0.1 mm	6 000
	Medium	3.5 mm ± 0.1 mm	3 000
	Light	3.0 mm ± 0.1 mm	2 600

(f) Rejection and Retest: Refer to SS 316.16.

316.05 Chain Link Fabric and Accessories on Rock Slopes – Chain link fabric for protection on rock slopes shall be manufactured and tested in general conformity with SS 316.04 and to the following requirements:

(a) Fabric shall be to CAN/CGSB-138.1M designation of wire woven to 75 mm diamond mesh in rolls 3.0 m wide, knuckled on both selvage edges, and 15 m long. See Table 316-E.

(b) For corrosive conditions and environment Grade 2 – heavy galvanizing may be specified as a special order with possible extended lead time required for delivery.

Table 316-E: Fabric Designation

Type 1	Class A Grade 1	Style	Nominal Diameter
Steel	zinc coated (galvanized)	Light	3.0 mm
		Medium	3.5 mm

316.06 Barbed Wire – Barbed wire for Type B & C fences and for security height Type D fences shall be manufactured from good quality galvanized steel wire in general accordance with ASTM A121 but shall conform to the following detail requirements:

(a) Wire Size and Strength – The wire shall be continuously twisted double strand of 2.5 mm nominal diameter (12 1/2 ga.) and to have a minimum breaking force of 4000 N.

(b) Barbs – Four point barbs shall be of not less than 2.0 mm nominal diameter (14 ga.) and regular 150 mm maximum spacings.

(c) Zinc Coating Class (according to ASTM A121) – The zinc used for coating shall be any grade of zinc conforming to ASTM B6. All wire, including barbs, shall have a uniform coating of zinc which shall adhere firmly to the base metal.

The average mass of zinc coating per unit area of uncoated wire for three specimens and the minimum for any individual specimens shall be in accordance with the class of coating as follows:

(i) Class 1 for use in the construction of Type C fences and for the top strand of Type B fences (see Drawings SP741-01.01 and SP741-02.01); the coating shall be average 90 g/m^2 and 80 g/m^2 individual minimums.

(ii) Class 3 for use with Security height Type D Chain Link fences (see Drawing SP741-05.01); coating shall be average 240 g/m^2 and 215 g/m^2 individual minimums.

(d) Length – Spools shall be 400 m unless otherwise ordered.

(e) Quality of Work – Splicing of individual strand wires by means of a wrap joint or an electric butt weld is permitted. Not more than three splices or joints shall exist in any 400 m spool of barbed wire. Such splices or joints shall be made in accordance with good work practice.

The twist in the double strand barbed wire shall be uniform and continuous in one direction to lock the barbs in place. The barbs shall be sharp and well formed.

Wire strands showing nicks or indentures likely to cause the wire to break under flexure are not acceptable.

(f) Testing

(i) Samples – Sufficient length of barbed wire shall be cut from each spool to be tested as required by SS 316.15.

(ii) Strength – The tensile strength of three line wire specimens shall be tested in accordance to ASTM A370 on a machine calibrated to ASTM E4 standards. The tensile strength shall meet the requirements of SS 316.06(a).

(iii) Coating Mass – Three single wire specimens (from each spool), each aggregating more than 400 mm in length shall be tested in accordance with ASTM A90 for conformity with the requirements of SS 316.06(c).

(iv) Rejection and Retest – Refer to SS 316.16.

Note: Smooth (barbless) galvanized wire, not specified to be "high-tensile," shall be double strand meeting all the above requirements except those for barbs.

Table 316-F: Requirements for Galvanized Wire for High Tensile Smooth-Wire (No Barbs) Fencing

Diameter:	2.5 mm nominal diameter (12 ½ ga.)
Tensile Strength:	between 1170 MPa and 1590 MPa (170,000 to 230,000 psi)
Breaking Force:	in excess of 5800 N (1300 lbf)
Zinc Coating (Class 3):	minimum 240 g/m^2
Coil Length:	in excess of 1200 m (approx. 100 lb.)

316.07 High-tensile Wire – Requirements for galvanized wire for high-tensile smooth-wire (no barbs) fencing are shown in Table 316-F.

316.08 Brace, Tension and Tie Wire

316.08.01 Brace Wires – Brace wire shall be manufactured of good commercial quality soft temper steel wire 3.5 mm nominal diameter gauge and zinc coated (galvanized) normally minimum 80 g/m^2 (Class 1 according to ASTM A641) or as otherwise specified or approved.

Brace wire for high-tensile smooth-wire fencing may be fence wire specified in SS 316.07.

316.08.02 Tension Wire – Tension wire to secure bottom of chain link fencing and at top where specified shall be manufactured of good commercial quality coil spring crimped steel wire 5.0 mm nominal diameter gauge and zinc-coated (galvanized) normally 400 g/m^2 (Class B according to ASTM A641) for Standard and Security height fences, or as otherwise specified or approved. Residential height fences tension wire shall be 3.5 mm nominal diameter.

316.08.03 316.08.03 Tie Wire and Wire Hog Ring Clips – Tie wire and wire hog ring clips shall be of steel of the nominal diameter gauge specified and zinc coated equivalent to that specified for the relevant fencing fabric: Aluminum tie wire shall be of specified gauge and according to ASTM B211.

316.09 Steel Fence Posts on Rock (Excluding Chain Link Fence) – Steel fence posts and braces for Type A, B and C fences where specified or required on exposed rock or rock with minimum over-burden shall be of good commercial quality steel conforming to either Category A or Category B as specified in SS 316.10 with the following exceptions:

(a) Line post mass shall be not less than 1.9 kg/m and angles 4.75 kg/m minimum, both drilled for attaching barbed wire or a substantial number of fence fabric line wires with wire ties or clips.

(b) Channel braces shall be 1.5 kg/m minimum.

316.10 Steel Pipe Posts, Rails and Braces – Chain Link Fencing

316.10.01 Category and Sizes – All posts, rails and braces shall be of Category 1 or Category 2 at the discretion of the supplier and shall be supplied in the nominal outside diameter as specified on the Purchase Order, Work Order or Special Provisions in accordance with the Drawings and/or Drawing SP741-05.02 and selected from Table 316-H or Table 316-G or as may be otherwise specified or approved.

Note: Specifications or Purchase Order may permit thin wall tubing and consequent lower zinc coating for proprietary galvanized and vinyl coated residential fence posts and gates.

Bolts and nuts for fastening braces to posts shall be galvanized Class D (305 g/m^2 individual pipe posts, rails and braces for chain link Type D fences shall be manufactured in general accordance with ASTM A53.

(a) Category A

(i) General – Category A posts, braces and rails shall be steel butt weld pipe, Schedule 40, as

Table 316-H: Requirements for Pipe Components

Actual Outside Post Diameter (mm)	26.7	33.4	42.1	48.3	60.3	73.0	88.9
Wall Thickness (mm)	2.87	3.38	3.56	3.68	3.91	5.16	5.49
Mass (kg/m)	1.7	2.5	3.4	4.0	5.4	8.6	11.3
Tolerances:	Diameter ± 1%			Mass ± 5%			

Table 316-G: Requirements for Pipe Components for Chain Link Fencing – Category A

Pipe Sizes	Nominal Outside Diameter (mm)		
	Residential	Standard	Security
Line Posts	42	48	60
End, Corner & Straining Posts	48	60	73
Gate Posts, gates 2.5 m and less	48	60	73
Gate Posts, gates over 2.5 m	48	73	89
Top Rails	33	33	42
Brace Rails	--	33	42

specified by ASTM A53, except that the hydrostatic testing requirement is waived, and the minimum weight shall be no less than 95% of the nominal weight.

(ii) **Coating** – Pipe shall be zinc-coated in accordance with ASTM A53. This ASTM Standard requires, among other things, an average zinc coating of 550 g/m^2 and a minimum zinc coating of 490 g/m^2.

(b) Category B

(i) **General –** Category B posts, braces and rails shall be made from steel pipes. The pipe shall be formed from steel strip conforming to ASTM A1011. The steel in the formed pipe shall have a minimum yield strength of 350 MPa (50,000 psi). The product of the yield strength and section modulus of the pipe shall be not less than that of pipe meeting the requirements for Category A posts and rails.

(ii) **Outer Surface –** The outer surface of Category B posts, braces and rails shall have all of the following protective coatings, applied in the following order:

(A) Hot dip zinc coating, conforming to ASTM B6, High Grade or Special High Grade, applied at an average weight of 305 g/m2 (1.0 oz/ft2) and at a minimum weight of 275 g/m2 (0.9 oz/ft2), as determined by the test method in ASTM A90.

(B) Chromate conversion coating, with a weight of 30 micrograms/sq. in. ±15 micrograms/sq.in., as determined by a strip and weight method for the zinc and chromate and by determining the percentage of each by atomic absorption spectrophotometer.

(C) Clear acrylic polyurethane coating with a thickness of 0.100 mm ± 0.025 mm (0.4 ± 0.1 mils).

(iii) **Coatings –** The combined outer coating shall have an ability to resist:

(A) 1000 hours of exposure to salt fog with a maximum of 5% red rust when tested in accordance with ASTM B117.

(B) 500 hours of exposure to 100% relative humidity without blistering and peeling when tested in accordance with ASTM D2247; and

(C) 500 hours of exposure in a weatherometer without film cracking of the clear coat when tested in accordance with ASTM G152, Type E or EH carbon arc, or ASTM G155, Type B or BH xenon arc.

(iv) **Inner Surface –** The inner surface of Category B, posts, braces and rails shall have either a zinc coating as per ASTM F1043, Type B, applied at an average rate of 305 g/m^2 and a minimum rate of 275 g/m^2; or an ASTM F1043, Type D zinc coating with a minimum zinc powder loading of 91% by weight and a minimum thickness of 0.075 mm (0.3 mils). The inner coating shall have an ability to resist 650 hours of exposure to salt fog with a maximum 5% red rust when tested in accordance with ASTM B117.

316.10.02 Quality of Work – The finished pipes shall be reasonably free from injurious defects. All burrs at ends of pipes shall be removed.

The zinc or zinc/chromate/acrylic coating shall be free from injurious defects and excessive roughness.

316.10.03 Testing

(a) **Samples –** A 100 mm specimen shall be cut from each end of one pipe in every lot of 500 or fraction thereof.

Note: Specimens may be taken from the extreme ends of pipe that has been cut to length after galvanizing; otherwise, specimens shall be taken at least 200 mm clear of each galvanized end build-up.

(b) **Dimensions** – Specimens shall be measured for diameter and wall thickness/mass and all shall be within the tolerance measurements specified by SS 316.10.01.

(c) **Mass Coating** – Specimens shall be tested for the mass of zinc or composite coating in accordance with SS 316.10.01 and ASTM A90.

(d) **Rejection and Retest –** Refer to SS 316.16.

316.11 Gates and Hardware – Gates shall be supplied with hardware ready for installation as specified by the Purchase Order, Work Order or Special Provisions in accordance with the Drawings and/or Drawings SP741-04.04, SP741-04.05, SP741-05.03 and SP741-05.04 or as may be otherwise specified or ordered.

Gates may be specified by the types as follows:

- **Type I** Pressed sheet steel frame, in-fill and brace members
- **Type II** Pipe frame and horizontal pipe in-fill members
- **Type III** Pipe frame, bracing and wire fabric in-fill
- **Type IV** Pipe frame, bracing and chain link fabric in-fill

- **Type V** Slip wire range gates barbed wire and wood droppers

with the general requirements of material, finish, hardware and quality of work specified in SS 316.11.01 through SS 316.11.06.

316.11.01 General – Gate sizes shall be the fence height or as otherwise stipulated and the nominal width designated by the clear opening width between gate posts with allowance made for hinges and latches, etc.

Gates shall be fabricated of galvanized pipe, tubing or formed sheet metal sections properly joined, braced and with in-fill all as specified by type below.

All pipe joints shall be cut to fit closely together and fully welded, vertical stiles neatly capped or mitred to horizontals with drilled holes to vent and drain closed end pipe members. Bends where permitted at corners shall be made without deforming tube or pipe section.

All welds, conforming to the best commercial practice, shall be smooth, and with adjoining and other marred galvanized pipe surfaces, shall be painted with two heavy coats of zinc rich paint selected from the Ministry's Recognized Products List under the category of "Additional Paint Coatings – Zinc-Rich Touch-up Paints and Primers".

All hardware and required fittings and fasteners shall be of suitable aluminum alloy or of steel or malleable or ductile cast iron, galvanized in accordance with ASTM A153. Hinges generally shall be so designed as to permit the gate to swing back against the fence and to prevent the easy removal of gate and hinges. Hinges and other fittings shall all be of adequate strength for fastening securely without distortion and shall be sound and free of defects. Chain and/or latching gate catch for padlocking together with keepers to hold the gate open and centre drop bolt and sleeve for double swing gates shall be provided as and where specified or ordered.

All galvanizing and other finished surfaces shall be smooth and free of sharp dangerous protrusions.

316.11.02 Type I Gate (sheet metal members) – Formed heavy duty galvanized sheet metal sections with rolled edges as horizontal rails and vertical stiles, braced vertically and/or diagonally with similar members as necessary shall be rigidly riveted together to form a rugged non-sagging prefabricated gate.

Nominal sizes:

- 1220, 2440, 3050, 3660, 4270 and 4880 mm wide; and
- 1220 mm high.

Gates to be supplied completely assembled with required hardware for wood (or metal) gate posts all as specified or ordered.

316.11.03 Type II Gates (all pipe) – Galvanized pipe 42 mm OD, 3.4 kg/m mass with zinc coating minimum 490 g/m^2 (see SS 316.10.01 and SS 316.10.03) shall be welded together as gate frame and horizontal pipe in-fill members at 305 mm on centres. Vertical centre bracing of 5 mm x 19 mm flat bar on both sides of all horizontals is not required for gates 3050 mm and 3660 mm nominal widths. Gussets shall be welded to opposite corners of the gate as shown on Drawing SP741-04.05.

Nominal sizes:

- 3050, 3660, 4270 and 4880 mm wide; and
- 1220 mm high.

Gates shall be supplied completely assembled with required hardware for wood (or metal) gate posts all as specified o ordered.

316.11.04 Type III Gates (pipe and fabric) – Galvanized pipe 33 mm OD, 2.5 kg/m mass with zinc coating minimum 490 g/m^2 (see SS 316.10.01 and SS 316.10.03) shall be formed with even radius bent corners or neatly welded corners and vertical bracing at 1/3 points in the width of 27 mm OD, 1.7 kg/m pipe. Cleats or gussets shall be welded to opposite corners of gate frame/panel(s) as indicated on SS Drawing SP741-04.05, holed for securing under tension a diagonal brace consisting of 5.0 mm nominal diameter galvanized coil spring wire (see SS 316.08) with or without forged turn-buckle as and when specified or ordered.

In-fill shall be wire fabric all of 3.5 mm galvanized diameter wire with seven or more horizontal line wires, and stay wires not more than 600 mm apart. Fabric shall be securely fixed taut without sag to gate frame all round and to vertical brace in an approved manner.

Nominal sizes:

- 3050, 3660, 4270, 4880 and 5490 mm wide; and
- 1070 and 1220 mm high.

Gates shall be supplied completely assembled with required hardware for wood (or metal) gate posts all as specified or ordered.

316.11.05 Type IV Gates (pipe and chain link) – Gate frame and required vertical bracing and/or diagonal bracing to cleats all welded as generally indicated on SS Drawing SP741-05.04 for hinged swing gates shall be fabricated to the designated size(s) complete with chain link in-fill and hardware for fixing to metal posts all as detailed, specified or ordered.

Pipe shall be in accordance with SS 316.10 and welding with SS 316.11.01. In-fill of chain link fabric as specified for fencing (see SS 316.04), shall be attached to gate frame stiles with tension bars and bands (see SS 316.12), and to horizontal members and vertical bracing with tie wire (see SS 316.08.03), all taut and sag free.

Sliding gates may be specified for gate openings over 4.0 m and shall be fabricated as generally indicated for swing gates but with diagonal tension rod bracing to support centre from both ends. The rolling hardware shall hold the gate vertical and parallel to the gate posts and adjoining fence at all times to allow free operation from any position while not retained by latches or keepers.

Roller assembly sheaves and wheels with bearings, grease fittings, necessary clamps, supports and fasteners, and chain and/or latching gate catch for padlocking, all of approved design, shall be provided as and when specified or ordered.

316.11.06 Type V Gates (range slip wire) – Barbed wire continued across standard width gate opening (for possible future farm gate) with 50 mm diameter full fence height treated wood droppers securely figure of eight wire tied and stapled to each line wire shall be provided with a galvanized steel strap loop at bottom and either similar loop at top with an intermediate chain (for padlocking) or, when specified, a Princeton or other approved cinch-up lockable latch device all for securely attaching to fence or gate posts to hold open-end dropper.

316.12 Accessory Material – All required fittings, fasteners, hardware and accessories shall be of suitable aluminum alloy or steel, malleable or ductile cast iron, and galvanized in accordance with ASTM A153, or as may be otherwise specified and/or ordered.

(a) Staples for securing wire and fabric to treated wood fence posts shall be nominal 3.5 mm diameter steel wire with a driving length of at least 45 mm heavy galvanized finish.

(b) Hog ring clips for securing wire fabric and/or barbed wire of Type A, B and C Fences to steel posts and to tension wire of Type D Fences see SS 316.08.03.

(c) Dowel, Spikes, Bolts and Nuts shall be of good commercial quality, galvanized where specified or ordered.

(d) Metal Droppers for high-tensile fencing will be prefabricated "Waratah" or other acceptable galvanized formed sheet metal droppers, 1070 mm for 5-wire and 1170 mm long for 6-wire fences, with preformed wire clips: such metal droppers or other acceptable manufacture may be specified or permitted as an alternative to the wood droppers specified by SS 909.08 for barbed wire (Type C) fences.

Note: Galvanized twisted double wire fence stays are permitted only for the extension of normal fence heights to deer height and the like.

(e) All terminal and tensioning posts, for Type D chain link fences, shall be provided with aluminum or galvanized weather tight caps to fit and fasten securely and with extended arms for engaging barbed wire of security fences.

(f) Caps for line posts for Type D chain link fences, shall carry top rail. Rail and brace end caps shall be provided for securing to rails and bolting with steel bands to posts.

(g) Tension Bars for securing chain link fencing to terminal and tensioning posts and to vertical stiles of gates of Type D Fences shall be merchant quality mild steel strip 5 mm x 19 mm in minimum lengths of 50 mm less than full height of the chain link fabric, hot dip galvanized.

(h) Tension and Brace Bands for securing brace and top rail caps and tension bars to posts shall be formed of minimum 2.5 mm x 19 mm mild steel strip, hot dip galvanized complete with bolt and nut.

(i) Tensioning Devices for individual in-line high-tensile wire will be "Hayes" or other acceptable manufacture and operation.

(j) Mechanical Splices and Fasteners shall be of "Nicopress FW2-3" or other acceptable manufacture applicable to the respective diameter and tensile strengths of wires.

316.13 Packaging and Marking – Each roll of fencing fabric or chain link mesh and spool of wire shall be tightly rolled and firmly tied to prevent loosening during shipment and handling.

Posts shall be securely fastened in bundles or sets, and accessories packaged in cartons or other suitable containers, to prevent loss or damage during shipment and handling.

Each product, roll, bundle or container etc., shall be clearly marked with the name, brand and/or trade mark of the manufacturer; the heat or lot number and other information sufficient to identify the style, type, class, grade, finish, height and/or length, nominal or outside diameter, gauge and the applicable Standard to which the product conforms.

316.14 Inspection – No material shall be shipped prior to inspection or until a release for shipment has been issued by the Ministry Representative.

All material may be inspected before shipment to the construction site or at the construction site, or both. The Supplier or manufacturer shall provide the necessary facilities to enable the Ministry Representative to expeditiously examine selected pieces from the material lot(s) clearly identified for intended supply to the Ministry. Identification shall be such that after inspection and testing the lot(s) may be subsequently identified for acceptance or rejection.

Inspection of material before shipment shall not exclude its subsequent rejection at the construction site if found to fail any requirements of this specification.

316.15 Sampling – One roll of fabric or spool of wire, in every lot of 50 or fraction thereof, in the shipment may be

selected at random for testing but in no case will less than two samples be tested.

Specimens for testing will be required from the outside ends of the selected sample roll or spool material and both ends of one post in every 500 or fraction thereof.

Sufficient length of each roll, spool or pipe shall be taken to provide the number of specimens for each of the testing requirements specified.

Wire used for knots, knuckling, twisting and wraps and wire containing welds shall be excluded from tests.

See respective material tests for pertinent sample lengths.

316.16 Rejection and Retests – If any tested specimen fails to meet the testing requirements specified above, the roll, spool or pipe represented by the specimens shall be replaced with two additional rolls, spools or pipes selected by the Ministry Representative and both shall meet the test requirements in every respect, otherwise the complete lot represented by the samples will be subject to rejection.

In the event of complete lot rejection and prior to any further testing by the Ministry, the intended replacement lot(s) shall be satisfactorily tested by a mutually acceptable independent testing agency at no expense to the Ministry: where the time delay for retesting is unacceptable to the Ministry the order or part thereof will be subject to cancellation.

Rejected material shall be replaced at the Supplier's or Contractor's expense including shipping charges and removal of rejected material at the construction site, if applicable.

316.17 Steel Sidewalk Fence – Steel pedestrian and bicyclist sidewalk fence material shall be installed in accordance with the requirements of the BC Supplement to TAC Geometric Design Guide, the Drawings and the details provided on SP741-07.01 and SP741-07.02. All steelwork shall be fabricated in accordance with SS 422. All field welded railing connections shall conform to the requirements of SS 422.

If the ‘swaged’ detail shown on SP741-07.03 is proposed as an alternative, the Contractor shall provide the Ministry Representative with satisfactory test results from an accredited testing agency for the following test procedures:

(a) Flattening test in accordance with ASTM A370;

(b) Whole body tensile test in accordance with ASTM A370; and

(c) Strain age embrittlement test in accordance with CSA C83 @–45ºC.

SECTION 317

P.E. PLASTIC DRAINAGE PIPE

DESCRIPTION

317.01 Scope – This specification covers the requirements for polyethylene plastic pipe products to be used for storm sewers, pipe culverts, and subdrains. This specification covers non-pressure polyethylene plastic pipe with nominal pipe diameters of 100 mm through to 900 mm. Larger pipe, from 900 mm up to 1500 mm, will be engineered (type, location application, backfill) as specified in the Special Provisions.

317.02 Applicable Documents – This specification refers to the following Standards, specifications or publications:

Canadian Standards Association (CSA):

B182.8: Profile Polyethylene (PE) Storm Sewer and Drainage Pipe and Fittings. (For nominal pipe diameters of 100 mm through to 900 mm.)

Note: Available only as part of the CSA B1800 Thermoplastic Non-Pressure Piping Compendium

American Society for Testing Materials (ASTM):

D883: Standard Terminology Relating to Plastics

D2122: Standard Test Method for Determining Dimensions of Thermoplastic Pipe and Fittings

D2412: Standard Test Method for Determination of External Loading Characteristics of Plastic Pipe by Parallel-Plate Loading

D2444: Standard Test Method for the Determination of Impact Resistance of Thermoplastic Pipe and Fittings by means of a Tup (Falling Weight)

D3212: Specifications for Joints for Drain and sewer plastic Pipes Using Flexible Elastomeric Seals

D3350: Standard Specification for Polyethylene Plastic Pipe and Fittings Materials

F667: Standard Specification for 3 Through 24 in. Corrugated Polyethylene Pipe and Fittings

American Association of State Highway and Transportation Officials (AASHTO):

M 252: Standard Specification for Corrugated Polyethylene Drainage Pipe. For nominal pipe diameter from 75 mm through to 250 mm.

M 294: Standard Specification for Corrugated Polyethylene Pipe. 300 to 1500 mm (12 to 60-in.) Diameter.

317.03 Definitions – For the purpose of this specification the following definitions apply.

(a) Polyethylene Plastic – A plastic based on polymers made with ethylene as essentially the sole monomer.

(b) Corrugated Inner Wall Pipe – A single walled pipe product with a corrugated inner waterway where the wall is formed into a series of alternating ridges and grooves.

(c) Profile – A pipe wall construction that presents an essentially smooth surface in the waterway but includes ribs or other shapes, which can be either solid or hollow, that help brace the pipe against diametrical deformation.

(d) Open Profile Pipe – A pipe product that has an essentially smooth waterway braced circumferentially or helically with outside corrugations, forming an open profile pipe.

(e) Closed Profile Pipe – A pipe that has a smooth waterway wall braced circumferentially or helically with corrugations, ribs, or other shapes that are joined integrally by an essentially smooth outer wall.

(f) Certification Body – An independent third-party agency, accredited by the Standards Council of Canada to confirm that a pipe manufacturer:

(i) Produces pipe products to the quality and requirements of an accepted CSA standard; and

(ii) Has the mandate to certify the pipe products produced.

(g) Certified – Pipe products that have been marked with a certification body's logo confirming that the production of the pipe product is in accordance with the quality and requirements of CSA B182.8.

317.04 Pipe Types and Materials – The pipe, fittings and couplers shall be manufactured from virgin polyethylene plastic, in accordance with the specifications listed in Table 317-A for the types and profiles of pipe.

317.05 Joining Systems

317.05.01 Flexible Subdrain Tubing – Couplings shall conform to the requirements of ASTM F667.

317.05.02 Flexible Culvert – Couplings shall conform to AASHTO M 252 or AASHTO M 294 Type C or CP.

317.05.03 Smooth Inner Wall Culvert or Storm Sewer – Joining system for open profile pipe shall conform to the requirements of CSA B182.8. Couplings shall be specified as non-gasketed corrugated external split couplings, or integral bell and gasket. Non-gasketed double bell snap

type couplings are only permitted for 100 mm to 200 mm diameters.

Integral bell and gasket joint shall be comprised of an elastomeric vulcanized rubber gasket provided on the spigot end of the pipe residing in the corrugation valley or crest of the corrugation. Joint shall be certified by CSA to meet leakage requirements of ASTM D3212.

Foam type gaskets shall not be permitted.

Table 317-A: Polyethylene (P.E.) Plastic Drain Pipe & Fittings Specifications

Type	Profile	Specification (See Note 1)
Flexible Subdrain Tubing	Corrugated Inner Wall	ASTM F667
Flexible Culvert	Corrugated Inner Wall	AASHTO M 252 or AASHTO M 294
Smooth Inner Wall Culvert or Storm Sewer	Open or Closed	CSA B182.8 for profile pipe

Note (1): Clean, rework PE material generated from the manufacturer's own production of pipe, fittings, or accessories shall be permitted to be recycled by the manufacturer in the respective product, provided that the pipe, fittings, or accessories produced meet all of the requirements of the respective standards. Post-consumer recycled plastic resin shall not be used as a component of the HDPE compound during pipe manufacture.

317.06 Requirements

317.06.01 Quality of Work – The pipe shall be homogenous throughout and free from visible cracks, flaws, foreign inclusions or other injurious defects. The pipe shall be uniform in colour, opacity and other physical properties.

317.06.02 Pipe Diameter

(a) Nominal Diameter – The average inside diameter rounded to the nearest whole number (mm).

(b) Inside Diameter – The permissible inside diameter tolerances shall be +3% and –1.5%. Measurements shall be taken in accordance with ASTM D2122.

317.06.03 Length – Laying length for flexible culvert; smooth inner wall culvert; or storm sewer shall be 6 m for all diameters otherwise specified on the Purchase Order, or Work Order or Drawings. A tolerance of ± 25 mm on the nominal laying length will be permitted.

317.06.04 Pipe Stiffness – The minimum pipe stiffness shall be:

(a) Flexible Subdrain Tubing – 210 kPa at 5% when tested in accordance with ASTM D2412.

(b) Flexible Culvert – 210 kPa at 5% when tested in accordance with ASTM D2412.

(c) Smooth Inner Wall Culvert or Storm Sewer – 320 kPa at 5% when tested in accordance with ASTM D2412. Pipe stiffness shall be documented on the Plant Certificate by a CSA accredited body for the appropriate pipe diameters.

317.07 Markings – All pipes shall be clearly marked in accordance with CSA B182.8, at intervals of no more than 1.5 m with 5 mm or larger letters with the following information:

- Manufacturer's name or trademark
- Nominal diameter
- Material designation and cell class: 322420C
- The words "Drainage" or similar
- Pipe stiffness (i.e. 320 kPa)
- The applicable Specification designation (i.e. CSA B182.8)
- Date of manufacture and plant designation
- Logo of accepted CSA third party Certified Agency

For Smooth Inner Wall Culvert or Storm Sewer, the pipe shall be embossed with the CSA Trademark of accepted third party confirming the product is certified by the Canadian Standards Council (CSC).

317.08 Quality Assurance Testing

317.08.01 Supplier Quality Control – The supplier shall develop and maintain an effective Quality Control system to ensure that adequate inspection coverage is maintained throughout the manufacturing process. Evidence of such inspection shall be available to the authorized inspector of the Ministry prior to shipment from the place of manufacture or the supplier's storage facility.

317.08.02 Ministry Surveillance – The Ministry reserves the right to maintain surveillance over the Supplier's Quality Control system to ensure conformance with this specification.

317.09 Physical Properties – The physical properties of corrugated HDPE pipe such as impact strength, pipe flattening and other tests shall meet these requirements for:

(a) Flexible Subdrain Tubing – ASTM F667

(b) Flexible Culvert – AASHTO M 252 or AASHTO M 294

(c) Smooth Inner Wall Culvert or Storm Sewer – Certified to CSA B182.8 for profile pipe.

317.10 Certificate

317.10.01 Flexible Subdrain Tubing and Flexible Culvert – Suppliers shall provide a certificate upon request from the Ministry Representative to indicate that the product was produced and tested according to the appropriate specification requirements.

317.10.02 Smooth Inner Wall Culvert or Storm Sewer – The supplier shall provide, upon request from the Ministry Representative, a copy of the CSA certificate of compliance issued by a CSA accredited certification body confirming that the manufacturer produces certified polyethylene plastic pipe products and indicating plant locations, pipe sizes, pipe stiffness and pipe joining systems.

317.11 Inspection

317.11.01 317.11.01 Ministry Access – The Ministry Representative shall have access to the fabrication plant for inspection and every facility shall be extended for this purpose.

317.11.02 Inspection – Inspection may include checks on physical dimensions and general quality of work.

317.11.03 Sampling – For the purpose of sampling for SS 317.06 and SS 317.09, one piece from each lot of 50 pipes or fraction thereof in a shipment shall be selected for sampling. All test specimens shall be cut from the selected pipe(s).

317.11.04 Failed Sample – If the test sample fails to meet the requirements of SS 317.06 when tested in accordance with SS 317.09, that batch will be rejected and shall be replaced by the supplier with a new batch that conforms to this Specification at no additional cost to the Ministry.

317.11.05 Acceptance – Pipes shall be made available for inspection and acceptance at the point and time of incorporation into the Work.

SECTION 318
PVC PLASTIC DRAINAGE PIPE

DESCRIPTION

318.01 Scope – This Section covers the requirements and methods for testing for:

- Type PSM* Poly Vinyl Chloride Drainage Pipe
- Profile Poly Vinyl Chloride Drainage Pipe

and the perforation requirements of type PSM pipe up to a nominal diameter of 375 mm.

Note: The term PSM is not an abbreviation but rather an arbitrary designation for a product having certain outside dimensions.*

318.02 Applicable Documents – This Section refers to the following standards, specifications and publications.

(a) Canadian Standards Association (CSA)

- B182.2 PSM Type Polyvinylchloride (PVC) Sewer Pipe and Fittings
- B182.4: Profile Polyvinylchloride (PVC) Sewer Pipe and Fittings

(The above are available only as part of the CSA B1800 Thermoplastic Non-Pressure Piping Compendium)

(b) American Society for Testing Materials (ASTM)

- D883: Standard Terminology Relating to Plastics
- D1784: Standard Specification for Rigid Poly(Vinyl Chloride) (PVC) Compounds and Chlorinated Poly(Vinyl Chloride) (CPVC)
- D2122: Standard Method of Determining Dimensions of Thermoplastic Pipe and Fittings
- D2152: Standard Test Method for Adequacy of Fusion of Extruded Poly(Vinyl Chloride) (PVC) Pipe and Molded Fittings by Acetone Immersion
- D2412: Standard Test Method for Determination of External Loading Characteristics of Plastic Pipe by Parallel-Plate Loading
- D2444: Standard Test Practice for Determination of the Impact Resistance of Thermo-plastic Pipe and Fittings by Means of a Tup (Falling Weight)
- D3034: Standard Specification for Type PSM Poly(Vinyl Chloride) (PVC) Sewer Pipe and Fittings
- D3212: Standard Specification for Joints for Drain and Sewer Plastic Pipes Using Flexible Elastomeric Seals
- F412: Standard Terminology Relating to Plastic Piping Systems
- F477: Standard Specification for Elastomeric Seals (Gaskets) for Joining Plastic Pipe
- F794: Standard Specification for Poly(Vinyl Chloride) (PVC) Profile Gravity Sewer Pipe and Fittings Based on Controlled Inside Diameter.

MATERIALS

318.03 Materials – The pipe shall be made of PVC plastic having a cell classification of 12454, or 12364, as defined in ASTM D1784.

The fittings shall be made of PVC plastic and shall have a cell classification of 12454, or 13343, as defined in ASTM D1784.

Elastomeric gaskets shall comply with the requirements of ASTM F477.

318.04 Quality of Work – The pipe shall be homogenous throughout and free from visible cracks, flaws, foreign inclusion, or other injurious defects. The pipe shall be uniform in colour, opacity and other physical properties.

318.05 Joining System – All sizes of pipe shall be supplied with Elastomeric Gasket Joints providing a watertight seal in accordance with ASTM D3034 and ASTM D3212.

318.05.01 Gasket Compression – Integral Bell Gasketed Joints shall be designed so that when assembled, the gasket shall be compressed radially on the pipe spigot or in the bell to form a watertight seal.

318.05.02 Gasket Displacement – The joint shall be designed to avoid displacement of the gasket, when assembled in accordance with the manufacturer's recommendation.

318.05.03 Joint Assembly – Joint assembly shall be in accordance with the pipe manufacturer's recommendations.

Note: Manufacturers wishing to supply pipe joining systems other than those listed under this specification shall submit details to the Ministry Representative for approval prior to the awarding of the contract.

318.06 Requirements

318.06.01 Length – Laying length shall be 4 m or 6 m for all diameters unless otherwise specified by the Purchase Order, Work Order or Drawings. A tolerance of ±25 mm on the nominal laying length will be permitted.

318.06.02 Type PSM, PVC Drainage Pipe Dimensions – The average outside diameter and the minimum wall thickness of Type PSM PVC drainage pipe shall be as specified in Table 318-A when measured in accordance with ASTM D2122.

318.06.03 Profile PVC Drainage Pipe Dimensions – The average inside diameter and minimum wall thickness of Profile PVC drainage pipe shall be as specified in Table 318-B when measured in accordance with ASTM D2122.

318.06.04 Perforations – Perforated pipe up to the nominal diameter of 37.5 mm shall be type PSM and perforated in accordance with Figure 318-1 and Table 318-C.

The perforations shall be 8 mm to 10 mm in diameter, circular, and cleanly cut. The gasketed bell and spigot end of the pipe shall be unperforated for a length equal to the depth of the socket and/or shoulder.

Table 318-A: Pipe Dimensions – Type PSM

Outside Diameter (mm)					
Nominal Diameter (mm)	**Average**	**Tolerance**	**Minimum**	**Maximum**	**Minimum Wall Thickness**
100	107.06	± 0.22	106.84	107.28	3.06
150	159.38	± 0.28	159.10	159.66	4.55
200	213.36	± 0.30	213.06	213.66	6.10
250	266.70	± 0.38	266.32	267.08	7.62
300	317.50	± 0.46	317.04	317.96	9.07
375	388.62	± 0.58	388.04	389.20	11.10
450	475.00	± 0.70	474.30	475.70	13.57
525	560.00	± 0.85	559.15	560.85	16.00
600	630.00	± 0.95	629.05	630.95	18.00
675	710.00	± 1.05	708.95	711.05	20.29

Table 318-B: Pipe Dimensions – Profile

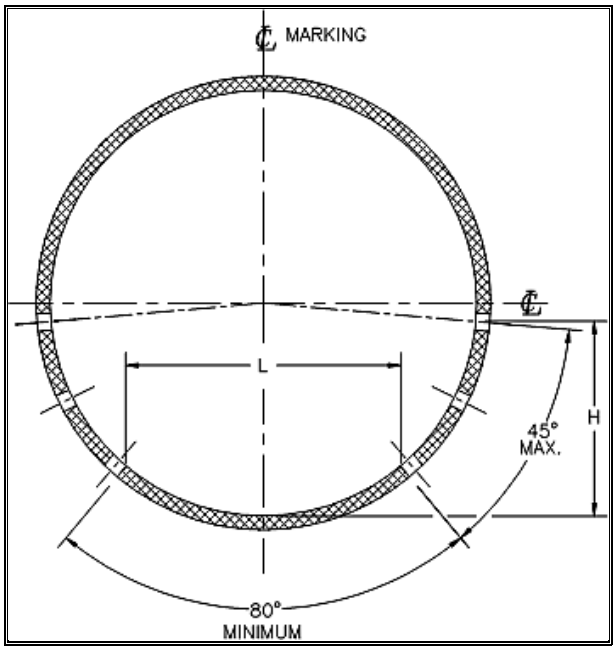
Inside Diameter (mm)						
Nominal Diameter (mm)	**Average**	**Tolerance**	**Minimum**	**Maximum**	**Dual Wall**	**Minimum Waterway Wall Thickness**
450	448.31	± 1.41	446.90	449.72	2.13	3.30
525	527.05	± 1.50	525.55	528.55	2.41	4.06
600	596.90	± 1.78	595.12	598.68	2.79	4.58
675	673.10	± 2.00	671.10	675.10	3.05	5.20
750	749.30	± 2.30	747.00	751.60	3.3	5.96
900	901.70	± 2.70	899.00	904.40	3.94	7.36
1000	1003.30	± 3.00	1000.30	1006.30	5.08	8.26
1200	1206.50	± 3.55	1202.95	1210.95	5.08	10.16

Table 318-C: Perforation Requirements

Nominal Diameter (mm)	Minimum Rows of Perforations	Horizontal Hole Spacing (mm)	H* (mm)*	L** (mm)
100	2	75 ± 5	30	65
150	4	75 ± 5	70	73
200	4	75 ± 5	93	130
250	4	75 ± 5	115	163
300	6	150 ± 5	140	195
375 - 1200	6	150 ± 5	175	240

**H: maximum height of the centreline of the uppermost perforations above the Invert.*
***L: minimum length of the unperforated inner surface at the bottom of the pipe.*

Figure 318-1: Perforated Pipe Cross Section



318.07 Service Connections (and or Branches)

318.07.01 Type PSM

(a) Injection Molded PVC Gasketed Fittings meeting ASTM D3034 and certified to CSA B182.2.

(b) Fabricated PVC Gasketed Fittings meeting ASTM D3034 and certified to CSA B182.2.

(c) Insertable Tees complete with rubber boot.

318.07.02 Profile

(a) Fabricated PVC Gasketed Fitting meeting ASTM F794 and Certified to CSA B182.4

(b) Insertable Tees complete with rubber boot.

318.08 Physical Properties

318.08.01 Impact Strength – The minimum drop weight impact strength for pipe shall meet the requirements of Table 318-D when tested in accordance with SS 318.10.01.

318.08.02 Pipe Stiffness – The minimum pipe stiffness when tested in accordance with SS 318.10.02 and ASTM D2412 shall be 320 kPa.

318.08.03 Pipe Flattening – There shall be no evidence of splitting, cracking or breaking when the pipe is tested in accordance with SS 318.10.03.

318.08.04 Extrusion Quality – The pipe shall not flake or disintegrate when tested in accordance with ASTM D2152.

Table 318-D: Impact Strength at 0ºC

Nominal Diameter (mm)	joules	ft. / lb.
100	135	100
150	165	122
200	175	129
250	190	140
300	205	151
375	230	170
450 and larger	250	184

318.09 Marking – All pipe shall be clearly marked at intervals of no more than 1.5 m with 5 mm or larger letters with the following information:

- Manufacturer's name or trademark
- Nominal diameter
- Material designation and cell class: PVC-12454 or 12364

- The word "Drainage" or similar
- The class: PS320
- The applicable specification designation (ASTM D3034 or ASTM F794)
- Date of manufacture and plant designation

The marking on perforated pipe shall be 180º from a point equidistant between the bottom row of holes as shown in Figure 318-1 and Table 318-C.

318.10 Test Methods

318.10.01 Impact Resistance – Five specimens, each nominally 150 mm long for the PSM pipe and 300 mm long for the Profile pipe, shall be tested in accordance with ASTM D2444, using a 15 kg Tup "A" and the flat plate holder "B". The height of the drop shall not be less than 600 mm or more than 3 m. The test shall be performed at 0ºC (± 0.5ºC), allowing no more than 15 seconds to complete the test after removing the sample from the conditioning medium. Perforated samples shall be positioned as shown in Figure 318-1.

Any cracking or other signs of material fatigue evident after the test shall constitute a failure.

If all five (5) specimens pass, the lot is acceptable.

If one specimen fails, five more specimens shall be tested.

If two or more specimens in the initial five (5) fail, or if any specimen in the second five (5) fails, the entire lot is rejected.

318.10.02 Pipe Stiffness – The pipe stiffness shall be determined at 5% deflection of the initial inside diameter in accordance with ASTM D2412. Three specimens, each with a length equal to one pipe diameter shall be tested; all shall meet the required stiffness. Perforated samples shall be positioned as shown in Figure 318-1.

318.10.03 Pipe Flattening – Three specimens, each nominally 150 mm long, shall be flattened between parallel plates in a suitable press until the distance between the plates is 40% of the original nominal diameter of the pipe. The loading rate shall be uniform and such that the compression is completed within two to five minutes. Remove the load and examine the specimen. All three specimens shall meet the requirements.

318.10.04 Pipe Dimensions – Measurements shall be taken in accordance with ASTM D2122.

318.11 Inspection

318.11.01 Access – The Ministry shall have access to the fabrication plant for inspection and every facility shall be extended for this purpose. For materials manufactured outside of British Columbia, SS 145.15.02 also applies.

318.11.02 Inspection – Inspection may include checks on physical dimensions and the quality of work.

318.11.03 Sampling – For the purpose of sampling for SS 318.08 and SS 318.10, one pipe from each lot of 50 pipes or fraction thereof in a shipment shall be selected for sampling. All test specimens shall be cut from the selected pipe(s).

318.11.04 Failed Sample – If a test sample fails to meet the requirements of SS 318.08 when tested according to SS 318.10, that batch will be rejected and shall be replaced by the Supplier with a new batch that conforms to this specification at no additional cost to the Ministry.

318.11.05 Acceptance – Pipes shall be made available for inspection and acceptance at the point and time of incorporation into the Work.

318.12 Quality Assurance

318.12.01 Supplier Quality Control – The Supplier shall develop and maintain an effective Quality Control system to ensure that adequate inspection coverage is maintained throughout the manufacturing process. Evidence of such inspection shall be available to the authorized inspector of the Ministry prior to shipment from the place of manufacture or from the Supplier's storage facility.

318.12.02 Ministry Surveillance – The Ministry reserves the right to maintain surveillance over the Supplier's Quality Control system to ensure conformance with this Specification.

SECTION 320
CORRUGATED STEEL PIPE

DESCRIPTION

320.01 General – This Section covers the material and fabrication requirements for corrugated steel pipe, spiral rib pipe, and structural plate corrugated steel pipe products for applications such as culverts, storm sewers, sanitary sewers, subdrains, ground recharge systems, well casings, underpasses, stream enclosures, shelters, and tunnels.

Abbreviations for the various types of steel pipe are as follows:

CSP Corrugated Steel Pipe means Galvanized, Aluminized Type II and Polymer Laminated corrugated steel pipe with helical corrugations, fabricated from coiled steel sheet, with continuous helical seam.

SPCSP Structural Plate Corrugated Steel Pipe means hot-rolled sheets or plates that are corrugated, curved to radius, custom hot-dip galvanized or has a thermoplastic copolymer coating, assembled, and bolted together to form pipes, pipe-arches, and other shapes. This includes DCSP Deep Corrugated Structural Plate (Type I, II, III).

320.02 Supply and Fabrication

320.02.01 Standards – The supply and fabrication of Corrugated Steel Pipe (CSP) and Structural Plate Corrugated Steel Pipe (SPCSP) including couplers and appurtenances shall be in accordance with CSA G401.

320.02.02 Plant Certification – All CSP and SPCSP shall be supplied from a manufacturing plant certified to CSA G401.

The certification shall be performed by a 3rd party agency accredited by the Standards Council of Canada. Certified CSP and SPCSP shall be marked according to CSA G401 markings, along with the logo of the 3rd party certification body.

The manufacturer's plant certificate documentation shall be made available to the Ministry upon request. This specification is available from CSA.

320.03 Quality of Work

320.03.01 Ministry Inspection – Products shall be made available for inspection and acceptance by the Ministry at the point and time of incorporation into the Work.

320.03.02 Quality Management – The manufacturer shall establish the Quality Control plan and demonstrate its implementation. The Contractor's Quality Management Plan shall incorporate Quality Assurance testing measures to assure the quality of the materials supplied by the manufacturer meet this Specification.

320.03.03 Contractor Quality Assurance Testing – Minimum Quality Assurance (QA) tests and inspection, to be performed by the Contractor, shall be per Table 320-A (next page).

Table 320-A: Contractor Quality Assurance Testing

Element	Description	Minimum Contractor Quality Assurance Testing / Inspection Frequencies	CSA G401 Clause No.
Manufacturer's QC	Demonstration of effectiveness	For each Designated Supplier (for over $200,000): Once per Contract	7
Material	Mill Certificate	One (1) per Contract	4.2.3; 4.3.3; 9.3; 4.1.2
	Mill Markings	One (1) per Contract and one additional per load of material delivered	4.5.1.2.4; 4.5.2.2.4; 4.5.3.2.4 4.2.3, 4.3.3
Fabrication	Thickness	One (1) per Contract per product	**CSP**: Table 3 **SPCSP**: Table 4
	Corrugation profile	Visual each piece, plus measurements of one (1) per Contract and one (1) additional per load of material delivered	**CSP:** 5.1.1.2.1, 5.1.1.2.3, Figure 4 and Table 6 **SPCSP:** 5.2.2.2 and Figure 12
	Circumference	One (1) per Contract per product	**CSP:** Tables 6, 7, 10, 11
	Lock seam strength	For Contracts with 10 or more pieces supplied: One (1) per Contract	5.1.3.4
	Perforation size	One (1) per shipment	5.1.5
Coating	Type	Visual, each pipe or plate	
	Thickness	For each material coating specified: One (1) per Contract and one additional per load of material delivered	4.5
	Staining or Damage	Visual, each piece	8.2
	Damage Repair	Each	6
Couplers	Type	Visual, each piece	5.1.7
SPCSP	Drawing and Plate Identification	Each installation	5.2.7

SECTION 321

TRAFFIC MARKING PAINT AND SPECIAL MARKINGS

DESCRIPTION

321.01 Scope – This Section applies to the supply and application of Traffic Marking Paint and Special Markings on concrete or asphalt roadway surfaces under the jurisdiction of the Ministry within the Province of British Columbia. Traffic Marking Paint (Paint) and Special Markings (Special Markings) shall be used with accepted overlay glass reflectorized beads.

321.02 Reference Specifications – The test methods, practices, and specifications listed in Table 321-A are applicable to the Work:

Table 321-A: Applicable Specifications

ASTM Test Designation	**Title of Test**
D562	Standard Test Method for Consistency of Paints Measuring Krebs Unit (KU) Viscosity Using a Stormer-Type Viscometer
D711	Standard Test Method for No-Pick-Up Time of Traffic Paint
D868	Standard Practice for Determination of Degree of Bleeding of Traffic Paint
D913	Standard Practice for Evaluating Degree of Traffic Marking Line Wear
D969	Standard Test Method for Laboratory Determination of Degree of Bleeding of Traffic Paint
D4061	Standard Test Method for Retroreflectance of Horizontal Coatings
D6628	Standard Specification for Color of Pavement Marking Materials
D7585	Standard Practice for Evaluating Retroreflective Pavement Markings Using Portable Hand-Operated Instruments
E1710	Standard Test Method for Measurement of Retroreflective Pavement Marking Materials with CEN-Prescribed Geometry Using a Portable Retroreflectometer
SAE Test Designation	**Title of Test**
SAE AMS-STD-595A	Standard Colors Used in U.S. Government Procurement

321.03 General Requirements – The composition and consistency of the Paint will be at the discretion of the manufacturer. Paint shall be supplied in white or yellow colours accepted by the Ministry without any "pre-mix" beads. Application of Paint shall include an overlay of glass reflectrorized beads.

Special Markings shall have reflective characteristics to meet Ministry requirements. Paint and Special Markings products shall meet Environment Canada's Federal Regulations on VOC's limits.

Special Markings material and application requirements shall be identified in the Special Provisions.

Traffic Marking Paint shall be supplied from traffic paint products listed on the Ministry's Recognized Products List, or as identified in the Special Provisions.

Special Marking Materials shall be supplied from the Ministry's Recognized Products List or as identified in the Special Provisions.

The Ministry's Recognized Product List is available on-line at:

https://www2.gov.bc.ca/assets/gov/driving-and-transportation/transportation-infrastructure/engineering-standards-and-guidelines/recognized-products-list/recognized_products_list.pdf

MATERIALS

321.04 Material Requirements – Paint and glass beads shall comply with the following:

321.04.01 Consistency – 85 ± 5 Krebs units in accordance with test method ASTM D562.

321.04.02 Drying Time – Maximum 30 minutes at 21ºC at 50-60% relative humidity in accordance with test method ASTM D711.

321.04.03 Bleeding – The Paint shall have a degree of resistance to bleeding of seven or greater in accordance with test method ASTM D868 or ASTM D969.

321.04.04 Road Service Life – Paint shall have acceptable properties for general appearance, durability and night visibility. Acceptable properties are defined as being equal to or better than those obtained by Paints currently "Qualified or Accepted" by the Ministry through Ministry product testing or other jurisdiction testing acceptable by the Ministry Representative.

321.04.05 Night-time Retroreflectance – Paint shall meet retroreflectance as per ASTM D7585 with the exceptions as identified within this Specification and the Ministry's Pavement Marking Contract. Retroreflectance shall be measured in accordance with ASTM E1710.

321.04.06 Application Characteristics

(a) The degree of settling shall be such that the Paint can be readily re-mixed to a uniform consistency and there shall be no skinning. Paint shall not have marked thixotropic properties.

(b) Paint shall be capable of being sufficiently atomized to produce a uniformly applied line 100 mm in width with absence of side splatter, overspray, or cobwebbing within the limits imposed by the application equipment used on the Contractor's line striping machines.

(c) Paint shall be capable of being successfully applied to at least 2 km of dash line without the necessity of making adjustments to any machine settings and without showing any evidence of distortion of the spray pattern or appreciable build up of paint in the spray gun tips.

321.04.07 Colour – In accordance to ASTM D6628:

(a) White Paint shall comply with SAE AMS-STD-595A White 17886.

(b) Yellow Paint shall comply with SAE AMS-STD-595A Yellow 33538

(c) Yellow Paint shall conform to the a-b colour box for traffic Paint using the L*a*b colour coordinate system.

321.04.08 Retroreflectance – Retroreflectance shall be in accordance with ASTM D4061.

321.04.09 Glass Beads – Glass reflectorized beads shall be added during the spraying of Paint in accordance with the glass beads manufacturer's recommended application requirements.

Glass beads shall be supplied from glass bead products listed on Ministry's Recognized Products List.

321.05 Traffic Markings: Paint shall be applied in the manner as required under the Work requirements. Types of Paint lines shall be identified by the Ministry in accordance to the "Manual of Standard Traffic Signs and Pavement Markings," Fig. 7.1. Ministry publications are available on-line at:

https://www2.gov.bc.ca/gov/content/transportation/transportation-infrastructure/engineering-standards-guidelines

Special Markings shall be laid out and identified in the Special Provisions.

PERFORMANCE SPECIFICATIONS

321.06 Performance Specifications

321.06.01 Longitudinal Pavement Markings – The Contractor shall:

(a) Layout all, with no additional payment, Longitudinal Pavement Markings as identified on the Drawings or existing pavement markings in accordance with the Ministry's "Manual of Standard Traffic Signs and Pavement Markings" or as instructed by the Ministry Representative.

(b) Paint Longitudinal Pavement Markings on New Pavement Projects as follows:

(i) Apply an initial application of Longitudinal Pavement Markings at a wet thickness of not less than 10 mils; then after curing;

(ii) Apply a second application, with no additional payment, in accordance with the paint manufacturer's suggested application interval. The wet thickness shall not be less than 15 mils regardless of product used; and

(iii) The wet thickness of all Longitudinal Pavement Markings will be applied uniformly.

(c) Ensure all Longitudinal Pavement Markings have well defined edges and are free of tire tracking, with no splatter, excessive overspray or other defects;

(d) Ensure Longitudinal Pavement Markings are straight, or of uniform curvature, and conform with the alignment;

(e) Ensure the following dimensional criteria are met:

(i) Longitudinal Pavement Markings do not exceed a dimensional width of 110 mm for specified 100 mm wide line. No tolerance below 100 mm is allowed for the specified 100 mm line;

(ii) Longitudinal Pavement Markings do not exceed a dimensional width of 210 mm for specified 200 mm wide line. No tolerance below 200 mm is allowed for the specified 200 mm line;

(iii) The distance between the lines for Double Solid and Simultaneous Solid and Broken do not exceed a dimensional separation of 110 mm. No tolerance below 100 mm is allowed;

(iv) Longitudinal Pavement Marking Broken Lines do not exceed a maximum dimensional length deviation of ±100 mm for the specified length; and

(v) No spaces between Broken Lines exceed a maximum dimensional length deviation of ±100 mm from the specified length of space.

(f) Apply glass beads immediately following the final Paint application. Glass beads are to be applied on all Longitudinal Pavement Markings at a uniform application rate as required by the glass bead Manufacture.

(g) Ensure:

(i) Retroreflectance properties of not less than 200 millicandela·m^{-2}·lux^{-1} for White Paint under dry conditions are achieved on all Longitudinal Pavement Markings for at least 30 days from the time of application;

(ii) Retroreflectance properties of not less than 150 millicandela·m^{-2}·lux^{-1} for Yellow Paint under dry conditions are achieved on all Longitudinal Pavement Markings for at least 30 days from the time of application;

(iii) Retroreflectance properties of not less than 150 millicandela·m^{-2}·lux^{-1} for White Paint under dry conditions are achieved from time of application to October 15th of the calendar year in which the line was painted;

(iv) Retroreflectance properties of not less than 100 millicandela·m^{-2}·lux^{-1} for Yellow Paint under dry conditions are achieved from time of application to October 15th of the calendar year in which the line was painted;

(v) All durability testing is in accordance with ASTM D913;

(vi) The condition of Longitudinal Pavement Markings shall be have a minimum of ninety-seven percent (97%) of the substrate covered, determined by interpolation from the ASTM D913 Photographic References, from time of application to October 15th of the calendar year in which the line was painted. Longitudinal Pavement Markings that do not meet this criteria must be repainted immediately upon detection or as directed by the Ministry Representative; and

(vii) The condition of Longitudinal Pavement Markings shall have a minimum of seventy-seven percent (77%) of the substrate covered, determined by interpolation from the ASTM D913 Photographic References, on January 15th of the subsequent year to which the line was painted.

321.06.02 Transverse, Chevron and Crosshatch Pavement Markings – The Contractor shall:

(a) Repaint the Transverse, Chevron and Crosshatch Pavement Markings, with no additional payment, as identified on the Drawings or existing pavement markings to conform to the Ministry's "Manual of Standard Traffic Signs and Pavement Markings." The Ministry Representative shall be contacted about any existing Transverse, Chevron and Crosshatch Pavement Markings that do not conform to the Ministry's "Manual of Standard Traffic Signs and Pavement Markings" prior to proceeding with the repainting;

(b) Layout and Paint all new Transverse, Chevron and Crosshatch Pavement Markings in accordance with the Ministry's "Manual of Standard Traffic Signs and Pavement Markings" or as instructed by the Ministry Representative;

(c) Apply Paint to Transverse, Chevron and Crosshatch Pavement Markings at a wet thickness of not less than 16 mils. This may be done in several coats as long as underlying coats are cured prior to overcoating;

(d) Ensure all Transverse, Chevron and Crosshatch Pavement Markings are free of tire tracking, with no splatter, excessive overspray or other defects;

(e) Ensure the ends of all Chevrons and Crosshatching are within 50 mm of the centre of the intersecting longitudinal line;

(f) Ensure all Transverse, Chevron and Crosshatch Pavement Markings have well defined edges and are free of horizontal fluctuations;

(g) Apply glass beads immediately following the final Paint application. Glass beads are to be applied on all Transverse, Chevron and Crosshatch Pavement Markings at a uniform application rate as required by the glass bead manufacturer; and

(h) Ensure:

(i) Retroreflectance properties of not less than 200 millicandela·m^{-2}·lux^{-1} for White Paint under dry conditions are achieved on all Transverse, Chevron and Crosshatch Pavement Markings for at least 30 days from the time of application;

(ii) Retroreflectance properties of not less than 150 millicandela·m^{-2}·lux^{-1} for Yellow Paint under dry conditions are achieved on all Transverse, Chevron and Crosshatch Pavement Markings for at least 30 days from the time of application,

(iii) All durability testing is in accordance with ASTM D913;

(iv) The condition of Transverse, Chevron and Crosshatch Pavement Markings shall have a minimum of ninety-seven percent (97%) of the substrate covered, determined by interpolation from the ASTM D913 Photographic References, from time of application to October 15th of the calendar year in which the marking was painted.

Transverse, Chevron and Crosshatch Pavement Markings that do not meet these criteria must be repainted immediately upon detection or as directed by the Ministry Representative; and

(v) The condition of Transverse, Chevron and Crosshatch Pavement Markings shall have a minimum of seventy-seven percent (77%) of the substrate covered, determined by interpolation from the ASTM D913 Photographic References, from time of application to June 15th of the subsequent year in which the marking was painted.

321.07 Quality Assurance – This section establishes how the pavement markings will be accepted or not accepted for conformance to the Specifications.

Test sites – Once per day that pavement markings are being laid, a minimum of two test sites are to be selected. The selected test sites must be representative of the line type sprayed on the pavement surface of that day (e.g., if 90% of the accomplishment of the day was centreline, than the expectation is that the test sites will be on centreline; if 50% of the accomplishment was on centreline and 50% was on lane line, then the expectation would be that there would be a test site on each line type).

321.07.01 Testing – The following tests must be performed and recorded per test site for that day:

(a) **Paint Colour** – Testing for conformance shall be made by visual comparison to a standard colour card provided by the Paint manufacturer that certifies colour compliance with this specification.

(b) **Dimensions** – Longitudinal Pavement Markings shall be measured and documented for conformance with the Pavement Marking Specifications.

(c) **Night-time Retroreflectance** – Retroreflectance will be measured as per ASTM D7585 and ASTM E1710, notwithstanding the following exceptions contrary to ASTM D7585:

(i) Contrary to Sections 6.2.1.4 and 6.2.2.4 of the ASTM D7585, checkpoint areas will be as per test sites, as detailed in SS 321.07;

(ii) Contrary to Sections 6.2.1.4 and 6.2.2.4 of the ASTM D7585, all measurements made within a single checkpoint area may be averaged and recorded as an average. Any average of the readings that do not meet these Pavement Marking Specifications will be determined as unacceptable work; and,

(iii) Contrary to Section 5.1 of the ASTM D7585, testing for Retroreflectance will be carried out only when the newly painted surface is clean, dry, free of all excess beads, and after 24 hours of Paint being applied.

(d) **Longitudinal Pavement Markings** – shall have a minimum initial coefficient of retroreflective luminance as indicated in these Pavement Marking Specifications.

321.07.02 Night-time Retroreflectance Measurements – All measurements for night-time retroreflectance will be made using calibrated equipment, capable of measuring retroreflectance in accordance with the applicable standards, as approved in writing by the Ministry. Measurements will be made using the retroreflectometer manufacturer's instructions for operation and procedures and will be made only by competent staff.

Sampling will be made using a sample size of 20 test point measurements at 5 m intervals on a single line, tabulated as an average with standard deviation and percent relative standard deviation, and minimum/maximum values.

All measurements and related data will be retained by the Contractor in an electronic format approved by the Ministry Representative and will be submitted to the Ministry Representative on request.

321.07.03 Test Report – A test report shall be produced for each checkpoint area and will include the following information:

(a) Test date;

(b) Average of the readings at each test location, expressed as millicandelas per square metre per lux ($mcd \cdot m^{-2} \cdot lx^{-1}$); the average of the readings will be reported for each traffic direction for centrelines;

(c) Geographical location of the test site, referenced by offsets from LKI/RFI landmarks;

(d) Identification of the pavement marking material tested: type, colour, date painted, and transverse location on road (line type);

(e) Identification of the instrument used;

(f) Value and date of standardization of the instrument standard panel used;

(g) Remarks concerning the overall condition of the line, such as rubber skid marks, carryover of asphalt, snow plow damage, and other factors that may affect the retroreflectance measurement;

(h) Ambient temperature; and

(i) Operator's name.

321.07.04 Thickness and Consistency – Longitudinal, Transverse, Chevron, and Crosshatch Pavement Markings shall have a sufficiently thick cross section throughout their entire length and width, and completely cover the intended area being marked. Pavement markings will be tested as per